PDE Ledger

Archive Ledger Skeleton

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Purpose and Scope

TODO. B3 will fill the rebuilt ledger purpose, source hierarchy, and scope boundaries.

0.1 The Role of This Ledger

Placeholder for the rebuilt ledger's role and audience.

0.2 Primary Source Hierarchy

Placeholder for the source hierarchy used by the rebuilt ledger.

0.3 Document Contract

Placeholder for the provenance, status, verification, and downstream-use contract.

0.4 Scope Boundaries

Placeholder for the limits of the rebuilt ledger.

0.5 Completeness Criterion

Placeholder for the completion standard used by later build phases.

How to Read This Ledger

TODO. B3 will fill the reader guide after the rebuilt stage and Part structure is set.

0.6 Layer Structure

Placeholder for the relationship between theorem-block prose, stage cards, notes, and audits.

0.7 Reading Paths

Placeholder for operational, verification, provenance, and authoring reading paths.

0.8 Citation Discipline

Placeholder for result anchors, theorem labels, equation labels, and stage provenance.

0.9 Stage Entries

Placeholder for the expected fields in future stage cards.

0.10 Gap Handling

Placeholder for demotion, splitting, open-gate, and failed-route handling.

Claim-Status Firewall

TODO. B3 will fill the rebuilt ledger's status rules and examples.

0.11 Allowed Status Tags

Status tag	Meaning
Exact	Placeholder.
Exact Within Closure	Placeholder.
Reduced / Controlled Reduction	Placeholder.
Effective Closure	Placeholder.
Numerically Located	Placeholder.
Open	Placeholder.

0.12 Audit-Status Vocabulary

Placeholder for audit evidence categories.

0.13 Status Inheritance

Placeholder for weakest-essential-input and status propagation rules.

0.14 Citation Language

Placeholder for safe status-aware citation wording.

Notation Firewall

TODO. B3 will fill notation rules after the rebuilt sectors and stage packets are fixed.

0.15 Global Anti-Ambiguity Rules

Placeholder for global notation safeguards.

0.16 Coordinate, Index, and Operator Conventions

Placeholder for coordinate, index, and operator conventions.

0.17 Field and Sector Variables

Placeholder for sector-specific variable conventions.

0.18 High-Risk Symbol Collision Table

Placeholder for symbol collision accounting.

0.19 Notation Checklist

Placeholder for future stage write-up checks.

Result Anchor Map

TODO. B3 will assign rebuilt result anchors after the Part and stage plan is settled.

0.20 Anchor Use Rules

Placeholder for stable citation-anchor rules.

0.21 Anchor Naming Convention

Placeholder for rebuilt anchor naming.

0.22 Top-Level Result Anchors

Anchor	Stage range	Citation target
No anchors assigned yet.		

0.23 Maintenance Checklist

Placeholder for anchor maintenance rules.

Dependency Map

TODO. B3 will fill the rebuilt dependency graph after stages are authored.

0.24 Arrow Types

Placeholder for dependency-arrow types.

0.25 Global Dependency Spine

Placeholder for the rebuilt ledger's dependency spine.

0.26 Top-Level Dependency Table

Block	Inputs	Outputs
No dependency blocks assigned yet.		

0.27 Revision Rules

Placeholder for dependency-map revision rules.

Contents

Part I

Part II – Gravity

Chapter 1

Gravity Ledger Stages

Stage cards for Part II are collected in Appendix A.

Appendix A

Part II Stage Cards

A.1 Stage 001: Solid-Angle and Second-Moment Primitives

Part / anchor. Part II – Gravity

Claim status. Exact

Purpose. Derive the unit-sphere solid angles and normalized second moments used by the pathA_21c matter-stress force assembly.

Status. Exact. This stage is closed geometry: no profile, normalization, or field-equation residual remains inside the stated claims.

Derivation. For $S^2 \subset \mathbb{R}^3$, take

$$n(\theta, \phi) = (\cos\theta, \sin\theta\cos\phi, \sin\theta\sin\phi), \qquad dS = \sin\theta\, d\phi\, d\theta.$$

Then **Omega2 solid angle.**

$$\Omega_2 = \int_0^{\pi} \int_0^{2\pi} \sin\theta\, d\phi\, d\theta = 4\pi.$$

With $n_1 = \cos\theta$ and $n_2 = \sin\theta\cos\phi$, **S2 second moments.**

$$\langle n_1^2 \rangle_{S^2} = \frac{1}{4\pi} \int_0^{\pi} \int_0^{2\pi} \cos^2\theta\, \sin\theta\, d\phi\, d\theta = \frac{1}{3}, \qquad \langle n_1 n_2 \rangle_{S^2} = 0.$$

The cross moment vanishes because $\int_0^{2\pi} \cos\phi\, d\phi = 0$.

For $S^3 \subset \mathbb{R}^4$, take

$$n(\chi, \theta, \phi) = (\cos\chi, \sin\chi\cos\theta, \sin\chi\sin\theta\cos\phi, \sin\chi\sin\theta\sin\phi), \qquad dS = \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi.$$

Then **Omega3 solid angle.**

$$\Omega_3 = \int_0^{\pi} \int_0^{\pi} \int_0^{2\pi} \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = 2\pi^2.$$

With $n_1 = \cos\chi$ and $n_2 = \sin\chi\cos\theta$, **S3 second moments.**

$$\langle n_1^2 \rangle_{S^3} = \frac{1}{2\pi^2} \int_0^{\pi} \int_0^{\pi} \int_0^{2\pi} \cos^2\chi\, \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = \frac{1}{4}, \qquad \langle n_1 n_2 \rangle_{S^3} = 0.$$

The cross moment vanishes because $\int_0^{\pi} \cos\theta\sin\theta\, d\theta = 0$.

Physical use. $\Omega_2 = 4\pi$ and $\Omega_3 = 2\pi^2$ set the reduced r^{-2} and bulk R^{-3} Gauss normalizations. The second moments give the δ_{ij}/d angular average used in the convective factor $-(1+1/d)$ and the pressure factor $1/d$.

A.2 Stage 002: Matter-Stress Inter-Defect Force Assembly

Part / anchor. Part II – Gravity

Claim status. Reduced / Controlled Reduction

Purpose. Assemble the inter-defect force between two drain defects from the matter (Noether) stress, earning the bilinear Q_1Q_2 structure, the reduced-3D r^{-2} and bulk-4D R^{-3} power laws, and the attractive like-drain sign, consuming the Stage 001 geometry primitives.

Status. Reduced / Controlled Reduction. Three verdict layers are carried from the earned source: the leading matter-stress sign `FORCE_ATTRACTIVE_DERIVED` (target-blind); the full sign `SIGN_RESIDUAL_QUANTUM_VCONF_MAXWELL_PROFILE` (honest residual); acceptance `PASS_WITH_NAMED_RESIDUALS`. The bilinear structure, the power laws, and the attractive sign are earned (form + sign); the overall magnitude is a calibration knob; the full sign carries named, profile/sim-deferred residuals.

Derivation. The medium obeys $P = K\rho^5$, $h = (5K/4)\rho^4$, with the barotropic identity $dP/d\rho = \rho\,(dh/d\rho)$. The matter (Noether) stress in Madelung form is

$$\Pi_{ij} = m\rho\, v_i v_j + \delta_{ij} P(\rho) + \sigma^Q_{ij}, \qquad \sigma^Q_{ij} = \frac{\hbar^2}{4m}\Big[\frac{(\partial_i\rho)(\partial_j\rho)}{\rho} - \partial_i\partial_j\rho\Big],$$

with the quantum-stress divergence contributing no net far-field force at this order. A defect draining flux Q sets up, by Gauss's law, the inflow

$$v = -\frac{Q\,\hat{n}}{\Omega_{d-1}\, r^{\,d-1}},$$

where $\Omega_2 = 4\pi$ $(d=3)$ and $\Omega_3 = 2\pi^2$ $(d=4)$ are the Stage 001 solid angles. The Bernoulli cross-pressure of the two overlapping inflows is $\delta P = -m\rho\,(v_1\cdot v_2)$. Integrating the traction $\Pi_{ij}n_j$ over a control sphere around defect 2 and using the Stage 001 second moment $\langle n_i n_j\rangle = \delta_{ij}/d$ gives the convective factor $-(1+1/d)$ and the pressure factor $+1/d$, whose sum is the total flux $-\,mNQ_2v_1$ (the $1/d$ pieces cancel). The stationary force $F_{12} = -\oint \Pi_{ij}n_j\, dS$, with defect 1's own inflow substituted, yields the two lanes.

Reduced-3D force law.

$$F_{12} = -\frac{m\,N_3\,Q_1Q_2}{4\pi\, r_{12}^2}\,\hat{r}_{12}.$$

Bulk-4D force law.

$$F_{12} = -\frac{m\,\rho_4\,Q_1Q_2}{2\pi^2\, R_{12}^3}\,\hat{R}_{12}.$$

Attractive sign. Since $F \propto -Q_1Q_2$, like drains $(Q_1Q_2 > 0)$ attract and opposite signs repel; the sign is derived from the Gauss-drain / Bernoulli / surface-integral chain, not assumed.

Physical use and named residuals. Target-blind, the inter-defect force is bilinear in the drain strengths, falls as r^{-2} (reduced 3D) and R^{-3} (bulk 4D), and is attractive between like drains. Honest scope: the overall normalization (I_F/Θ_Q/branch-profile) is a *calibration* knob, not derived; and the full sign carries the `QUANTUM_STRESS_FAR_FIELD`, `VCONF_BODY_FORCE`, and `MAXWELL_Z_GAUGE_JEXT` residuals, which are profile/sim-deferred and first-class.

Appendix B

Reproducibility Map

B.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's reproducibility rules.

B.2 Reproducibility Layers

Layer	What it controls	Minimum evidence required
No reproducibility layers populated yet.		

B.3 Canonical Source Registry

Placeholder for source artifacts.

B.4 Primary Symbolic-Audit Registry

Placeholder for symbolic audit artifacts.

B.5 Repository Freeze Protocol

Placeholder for future release-freeze requirements.

Appendix C

Source File Index

C.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's source-file index.

C.2 Source Classes

Placeholder for source classes.

C.3 Generated Ledger Files

File family	Role
`paper/frontmatter/*.tex`	Placeholder governance chapters.
`paper/parts/*.tex`	Future theorem-block chapters.
`paper/stages/stage_NNN.tex`	Future per-stage cards.
`scripts/ledger_stageNNN_*.py`	Future SymPy audit scripts.
`mathematica/ledger_stageNNN_*.wl`	Future Mathematica audit scripts.

C.4 Precedence and Conflict-Resolution Rules

Placeholder for source precedence and conflict-resolution rules.

Appendix D

Layer-2 Fit-Insertion Index

D.1 Generated Coverage Summary

Item	Count
Phase-A candidates	0
Audited candidates	0

D.2 Phase-A Candidate Rows

No candidate rows exist in the empty ledger skeleton.

D.3 Visible Grouping and Exclusion Accounting

No grouping or exclusion accounting exists yet.

Appendix E

Layer-2 Parameter-Provenance Audit

E.1 Generated Coverage Summary

Item	Count
Top-level provenance parameter records	0
Canonical candidates	0

E.2 Published-Target Benchmark Sources

No benchmark rows exist in the empty ledger skeleton.

E.3 Parameter Genealogy Rows

No parameter genealogy rows exist yet.

E.4 Visible Alias Accounting

No alias accounting exists yet.

Appendix F

Fill Workflow and Editorial Rules

F.1 Purpose of This Appendix

Placeholder for the rebuilt ledger fill workflow.

F.2 What Filled Means

Placeholder for completion criteria.

F.3 Governing Maps Used During Filling

Placeholder for status, notation, anchor, dependency, source, and reproducibility controls.

F.4 Stage Appendix Fill Protocol

Placeholder for future stage-card authoring rules.

F.5 Theorem-Block Fill Protocol

Placeholder for future Part authoring rules.

F.6 Minimal Build and Static-Check Workflow

1. Confirm every retained `\input` target exists.
2. Run the SymPy and Mathematica audit runners.
3. Build `paper/pde_ledger.tex`.